



SCHOOL OF DATA SCIENCE

Statistics for Data Analysis

Syllabus

UDACITY.COM

Overview

Learn core statistics for data analysis, including probability, regression, and experiment design. Build practical Python skills to analyze data and prepare for analyst and data scientist roles.

Nanodegree Program



Beginner



64 hours



5.0 (2 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Basic Python

Basic arithmetic

Elementary algebra

NumPy

You will also need to be able to communicate fluently and professionally in written and spoken English.

Skills You'll Learn

Descriptive Statistics

Basic descriptive statistics | Outlier analysis

Probability

Conditional probability | Probability distribution | Bayes' theorem | Basic probability

Hypothesis Testing

Normal distribution | Experiment design | Statistical sampling | Pandas | A/B testing | Inferential statistics | Interpreting test results

Linear and Logistic Regression

Statistical modeling fundamentals | Logistic regression | Linear regression | Linear functions

Courses

01. Welcome to the Statistics for Data Analysis Nanodegree Program

45 minutes

Welcome to Udacity! We're excited to share more about your Nanodegree program and start this journey with you!

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| L1 | An Introduction to Your Nanodegree Program | Welcome! We're so glad you're here. Join us in learning a bit more about what to expect and ways to succeed. |
| L2 | Getting Help | You are starting a challenging but rewarding journey! Take 5 minutes to read how to get help with projects and content. |

02. Descriptive Statistics

9 hours

Learn how to describe data in terms of data types, measures of center, measures of spread, shape, and outliers. These essential skills in descriptive statistics provide the foundation for more advanced statistical techniques that are used for data science, data analysis, and machine learning.

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| L1 | Introduction | In this lesson, we kick off the course with an introduction to data and descriptive statistics. |
| L2 | Data Types | In this lesson, we establish key distinctions between different types of data, including quantitative, categorical, ordinal, nominal, continuous, and discrete data types. |
| L3 | Measures of Center | In this lesson, we get into the calculations and use cases for three popular measures of center: mean, median, and mode. |
| L4 | Notation | In this lesson, we demystify the mathematical notation used for random variables, observed values, and aggregations. |
| L5 | Measures of Spread | In this lesson, we get into the calculations and use cases for several popular measures of spread, including five-number summaries, standard deviation, and variance. |
| L6 | Shape and Outliers | In this lesson, we cover two additional forms of descriptive statistics for data distributions: shape and outliers. |

- L7 Project:**
Describe Health and Sleep Quality Data

Describe data related to health and sleep quality. Using the software of your choice to calculate descriptive statistics about a dataset, you'll report your findings in a slide deck presentation.

03. Probability

13 hours

This course is a comprehensive dive into the fundamental concepts and principles of probability. You'll begin with basic probability theory, then progress to more complex topics such as binomial distributions, conditional probability, and Bayes' Rule. These skills will enhance your ability to reason about uncertainty and make claims using data.

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| L1 Introduction | In this lesson, we start off with meeting the instructors and a high-level overview of probability. |
| L2 Probability Basics | In this lesson, you'll gain the basics of probability using coins and dice. |
| L3 Binomial Distribution | In this lesson, you'll learn about one of the most popular distributions in probability - the Binomial Distribution. |
| L4 Conditional Probability | Not all events are independent. In this lesson, you'll learn the probability rules for dependent events. |
| L5 Bayes Rule | In this lesson, you'll learn one of the most popular rules in all of statistics - the Bayes Rule. |
| L6 Project:
Calculate Basketball Scoring Probabilities | Analyze data related to basketball shooting statistics. You'll use the software of your choice to calculate probabilities related to a dataset, then report your findings in a slide deck presentation. |

04. Hypothesis Testing

21 hours

Hypothesis testing is one of the most important topics in all of statistics because it tells us whether our conclusions are statistically significant. In this course, you will learn about the fundamental role statistics plays in hypothesis testing as well as how to implement statistical concepts in Python.

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| L1 Introduction | In this lesson, we begin the course by meeting the instructors and giving a quick introduction to experimentation. |
| L2 Normal Distribution Theory | In this lesson, you'll learn the mathematics behind moving from a coin flip to a normal distribution. |



L3	Sampling Distributions and the Central Limit Theorem	In this lesson, you'll learn all about the underpinning of confidence intervals and hypothesis testing - sampling distributions.
L4	Confidence Intervals	In this lesson, you'll learn how to use sampling distributions and bootstrapping to create a confidence interval for any parameter of interest.
L5	Hypothesis Testing	In this lesson, you'll learn the necessary skills to create and analyze the results of hypothesis testing.
L6	A/B Tests	In this lesson, you'll work through a case study of how A/B testing works in the context of website metrics for an online education company.
L7	Project: Analyze A/B Test Results	You will be working to understand the results of an A/B test run by an e-commerce website. Your goal is to work through to help the company understand if they should implement the new page design.

05. Linear and Logistic Regression

20 hours

The linear and logistic regression course offers a detailed introduction to fundamental statistical and machine learning algorithms, particularly focusing on regression techniques. The course begins with simple linear regression and progresses to multiple linear regression, equipping students with the ability to analyze relationships between multiple variables. Finally, it covers logistic regression, a powerful tool for classification problems.

L1	Introduction	In this lesson, we kick off the course with an introduction to machine learning and why linear regression is such an important algorithm.
L2	Regression	In this lesson, you'll use Python to fit linear regression models, as well as understand how to interpret the results of linear models.
L3	Multiple Linear Regression	In this lesson, you'll learn to apply multiple linear regression models in Python. Then you'll learn how to interpret the results and understand if your model fits well.
L4	Logistic Regression	In this lesson, you'll learn to apply logistic regression models in Python as well as how to interpret logistic regression results.
L5	Project: Model Salary Data with Linear Regression	In this project, you will build a linear model and interpret the results of the model. Specifically, you will look at what variables are related to salaries.

06. Congratulations!

10 minutes



Congratulations on finishing your program!

L1 Congratulations!

Congratulations on your graduation from this program! Please join us in celebrating your accomplishments.

Meet Your Instructors



Sebastian Thrun

Founder and Executive Chairman, Udacity

As the Founder and Chairman of Udacity, Sebastian's mission is to democratize education by providing lifelong learning to millions of students worldwide. He is also the founder of Google X, where he led projects including the Self-Driving Car, Google Glass, and more.



Josh Bernhard

Staff Data Scientist

Josh has been sharing his passion for data for over a decade. He's used data science for work ranging from cancer research to process automation. He recently has found a passion for solving data science problems within marketplace companies.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices